Date: 1st September 2026

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Information Security

Congruence

⇒ Defⁿ - Let a, b, n be integers with $n \neq 0$

$a \equiv b \pmod{n}$

(a is congruent to b mod n) if $a-b$ is a multiple of n.

$a \equiv b \pmod{n}$ if a and b differ by a multiple of n. So $a-b = nk$ or $a = b+nk$

Ex | $32 \equiv 2 \pmod{5}$, $17 \equiv 17 \pmod{13}$ | $-12 \equiv 37 \pmod{7}$ ⇒ $-5 \equiv 2 \pmod{7}$

Propositions

(i) $a \equiv 0 \pmod{n}$ iff n/a

(ii) $a \equiv a \pmod{n}$

(iii) $a \equiv b \pmod{n}$ iff $b \equiv a \pmod{n}$

(iv) If $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$ then $a \equiv c \pmod{n}$

Solⁿ (i) $a \equiv 0 \pmod{n}$ means $a = a-0$ is a multiple of n/a.

(ii) $a - a = 0 \cdot n$, for $k=0$

(iii) $b \equiv a \bmod n \rightarrow b-a = nk \Rightarrow a-b = n(-k)$

So $a \equiv b \pmod{n}$

(iv) $a = b + nk$ and $b = c + ln$

So,

$\Rightarrow a = c + ln + nk$

$a = c + (l+k)n$

$a \equiv c \pmod n$

Proposition

$a \equiv b \bmod n$, $c \equiv d \pmod n$

Then,

$a + c \equiv b + d \pmod n$

$a - c \equiv b - d \pmod n$

$ac \equiv bd \pmod n$

Addition over modulo n

Let $n = 6$ then addition modulo table is

+	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

Table for multiplication mod 6

X	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	1	2	3	4	5
2	0	2	4	0	2	4
3	0	3	0	3	0	3
4	0	4	2	0	4	2
5	0	5	4	3	2	1

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Ex - Solve $x+7 \equiv 3 \pmod{17}$

Sol^n $x \equiv 3-7 \pmod{17}$

$\Rightarrow x \equiv -4 \pmod{17}$

or $x \equiv 13 \pmod{17}$

Proposition

$\Rightarrow$ Let a, b, c, n be integers with $n \neq 0$ and with $\gcd(a, n) = 1$

If $ab \equiv ac \pmod{n}$, then $b \equiv c \pmod{n}$

In other words, if a and n are relatively prime we can divide both side of congruence by a.

Ex - Solve $2x+7 \equiv 3 \pmod{17}$

Sol^n $2x \equiv (3-7) \pmod{17}$

$2x \equiv -4 \pmod{17}$

Let $a = 2$, then

$\Rightarrow x = -2 \pmod{17}$

$x = 15 \pmod{17}$

Ex - Solve $5x+6 \equiv 13 \pmod{11}$

Soln $5x \equiv (13-6) \pmod{11}$

$5x \equiv 7 \pmod{11}$

~~Let $a \neq x$, then let~~

$7 \bmod 11 \Rightarrow 18 \equiv 29 \equiv 40 \pmod{11}$ (1)

Hence, this can be written as -

$5x \equiv 7 \bmod 11$

$5x \equiv 40 \bmod 11 \Rightarrow \boxed{x = 8 \bmod 11}$

(2nd Method)

Ex - $5x + 6 \equiv 13 \pmod{11}$

Soln $5x \equiv 7 \bmod 11$ — (1)

We Know,

$5 \cdot 9 \equiv 1 \pmod{11}$

Hence, inverse of 5 is 9,

that is, $\frac{1}{5} \rightarrow 9$

So, multiply by 9 on both ends in eqn (1)

$\Rightarrow 9 \cdot 5x \equiv 9 \cdot 7 \bmod 11$

$x \equiv 63 \bmod 11$

$x \equiv 8 \bmod 11$

Ques Solve $1111x \equiv 4 \pmod{12345}$

Date - 2 September 2026

Greatest Common Divisor

$\gcd(6,4) = 2$, $\gcd(5,7) = 1$

Method - 1

Ex - 576, 135

Soln $576 = 2^6 \times 3^2$

$135 = 3^5 \times 5$

So, $\gcd(576, 135) = 3^2$

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Method-2 (Using Euclidean Algo)

Q ≡ gcd (482, 1180)

Sol^n ≡

$1180 = 2 \times 482 + 216$
$482 = 2 \times 216 + 50$
$216 = 4 \times 50 + 16$
$50 = 3 \times 16 + 2$ ← Last non zero remainder
$16 = 2 \times 8 + 0$

Euclidean Algo

Let $a > b$

$a = q_1 b + r_1$

If $r_1 = 0$, then b divides a and gcd is b
If $r_1 \neq 0$ then continue

$a = q_1 b + r_1$
$b = q_2 r_1 + r_2$
$r_1 = q_3 r_2 + r_3$
$r_2 = q_4 r_3 + r_4$
⋮
⋮

$r_{k-2} = q_k r_{k-1} + r_k$
$r_{k-1} = q_{k+1} r_k + 0$

So $gcd(a, b) = r_k$

Extended Euclidean Algo

⇒ Given integers a and b, there are integers x and y such that

$ax + by = gcd(a, b)$

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Q Find integers x and y such that
gcd(56, 72) = 56x + 72y

$72 = 56 \times 1 + 16$

$56 = 16 \times 3 + 8$ — (1)

$16 = 8 \times 2 + 0$

gcd(56, 72) = 8

So, we can write eqn (1) as

$8 = 56 - (16 \times 3)$

$= 56 - [(72 - 56 \times 1) \times 3]$

$= 56 - [72 \times 3 - 56 \times 3]$

$= 56 - 72 \times 3 + 56 \times 3$

$= \underline{56 \times 4 - 72 \times 3}$

gcd(56, 72) = 56x + 72y

= 56(4) + 72(-3)

So, $\boxed{x = 4, \; y = -3}$

Q Find integers x and y such that
gcd(272, 1479) = 272x + 1479y